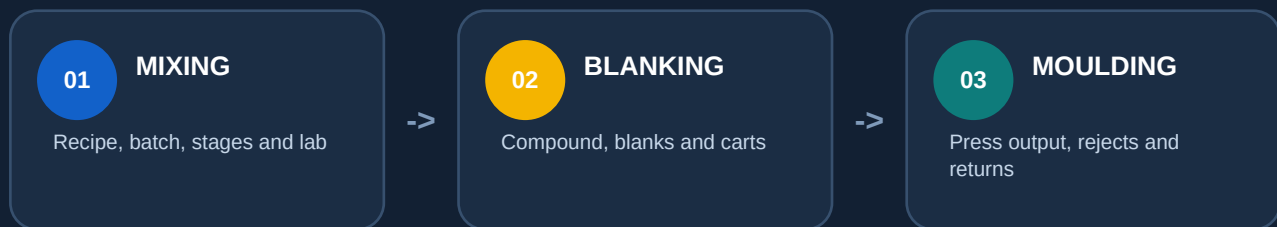


PROTOTYPE OVERVIEW

Real-Time Production Tracking System

A connected digital workflow for Mixing, Blanking and Moulding



Executive summary

The Stellana Real-Time Production Tracking System is a working factory prototype designed to replace fragmented paper and spreadsheet updates with one connected record. It gives authorized users a current view of production, preserves batch and recipe history, records material movement, supports operator updates, and creates traceable management reports. The architecture is suitable for desktop and industrial tablet use on the company network.

PROTOTYPE SCOPE

LIVE STATUS

RECIPE REVISIONS

BATCH TRACEABILITY

MATERIALS

LAB DECISIONS

CARTS AND PRESSES

AUDIT HISTORY

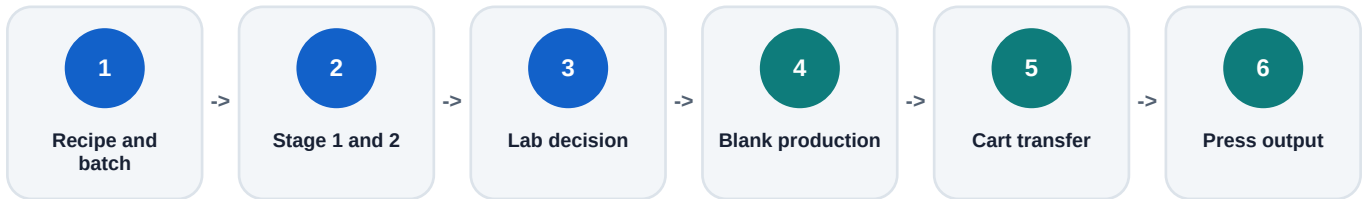
PDF REPORTS

CURRENT PROTOTYPE NOTE

Blanking currently accepts the physical Mixing batch number and compound code manually. Laboratory approval enforcement is temporarily relaxed until the factory Lab workflow is introduced.

What the system does

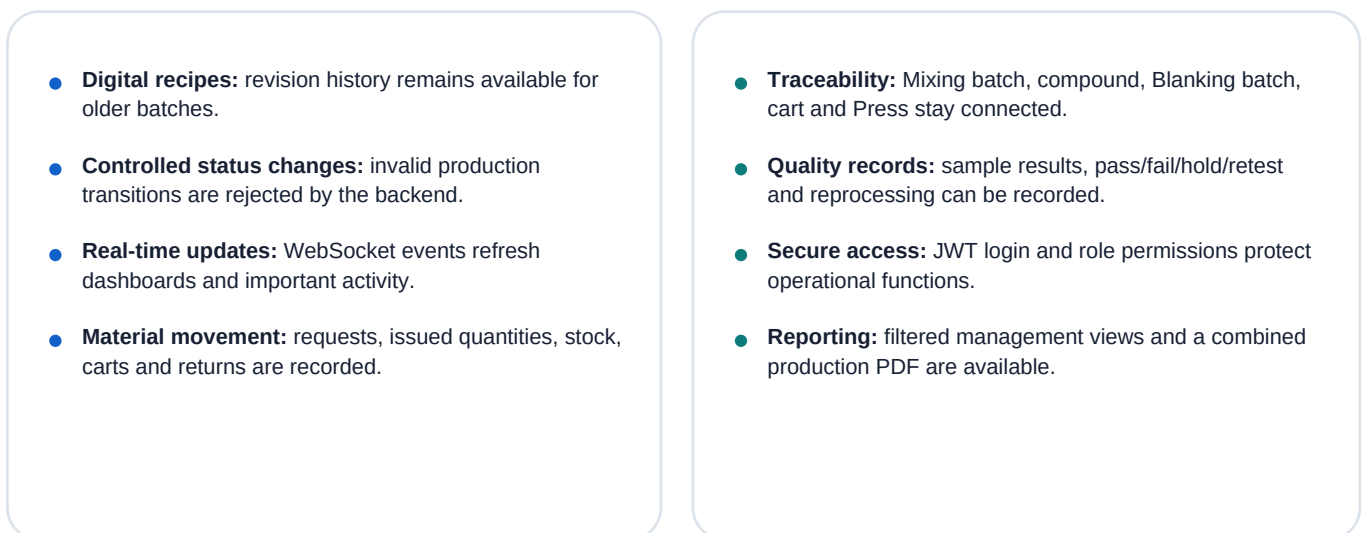
One digital chain follows production from the selected recipe revision to final press output and material return.



Role-based working views



Core capabilities in the prototype



ONE ENTRY - MANY USES

A production entry is reused for dashboards, batch history, traceability, inventory balances, notifications, audits and reports. This reduces repeated manual copying between papers and spreadsheets.

Advantages to Stellana

Expected operational benefits should be measured during a controlled factory pilot.

01

Real-time visibility

Managers can see active, waiting, stopped and completed work without calling or visiting each section.

02

End-to-end traceability

Production records remain connected from recipe revision and Mixing batch through carts and Press output.

03

Less paper and Excel

Operators enter data once, while reports and summaries reuse the same controlled records.

04

Stronger quality evidence

Sample decisions, test dates, reprocessing, release and history are stored against the correct batch.

05

Faster coordination

Live alerts, issue messages, shortage requests and cart status reduce uncertainty between departments.

06

More reliable reporting

Server timestamps, role permissions, validation and audit history improve confidence in management information.

From current difficulty to company value

CURRENT DIFFICULTY	SYSTEM RESPONSE	POTENTIAL VALUE
Status learned by calls or visits	Live dashboard and section events	Quicker awareness and escalation
Recipe papers can be separated from batches	Exact recipe revision stored on the batch	Improved production and quality traceability
Repeated spreadsheet preparation	Combined filtered report and PDF download	Less reporting effort and duplicate entry
Transfers and returns are hard to reconcile	Quantity ledger, cart receipt and return confirmation	Clearer stock responsibility and variance visibility

Suggested factory rollout

A controlled pilot can prove usability and data quality before the system becomes an official production record.

Simple company network setup



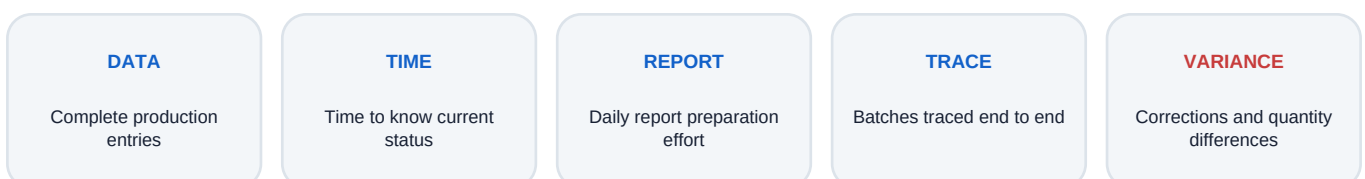
Recommended pilot steps

- **1. Prepare:** confirm server, PostgreSQL, backups and a reserved internal address.
- **2. Configure:** create real users, roles, machines and approved master data.
- **3. Pilot:** run one shift or selected product family with paper kept as backup.
- **4. Review:** compare quantities, timestamps and operator feedback every day.
- **5. Expand:** introduce official Lab and Stores roles after rules are approved.

Factory decisions still required

- Mixer fill-factor formula and allowed load
- Laboratory specifications and limits
- Recipe create, approve and activate responsibility
- Official Stores and Lab Officer responsibilities
- Rejection, hold, reprocessing and release approvals
- Backup, retention and disaster-recovery policy

Suggested pilot measures



CONCLUSION

The prototype provides a practical foundation for a shared production record across Mixing, Blanking and Moulding. Its main company value is visibility, traceability and reliable reuse of the same operational data. A supervised pilot should now validate the screens, responsibilities and factory rules before wider deployment.